IRT-FIT: Detecting Benchmark Contamination in LLMs from Binary Response Patterns Alone

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Abstract

Benchmark contamination threatens the reliability of LLM evaluation, yet every existing detection method requires privileged access beyond binary (correct/incorrect) responses—training data overlap, token probabilities, model weights, or specially crafted prompts. We introduce **IRT-FIT**, a contamination detection framework operating at the *minimal-information operating point* by repurposing Item Response Theory (IRT) person-fit statistics from psychometric cheating detection. Using controlled contamination experiments across four model families (Qwen3.5-9B, Gemma-4-31B, Llama-3.1-8B, Mistral-7B-v0.3) with QLoRA SFT injection at six contamination levels (0–50%) on three benchmarks, we show: (1) l_z^* with θ-fixed analysis detects contamination with $|d| \geq 2.42$ at 15% contamination across all four model families; (2) detection power follows a predictable function of benchmark size ($R^2 = 0.968$ across five benchmarks from 1.3K to 14K items); (3) after GRPO post-training, l_z^* retains 80% of its detection signal (AUROC 0.92→0.85) while token-level methods collapse to chance (Min-K%++ 0.89→0.54). At the practically critical 5% contamination level, l_z^* achieves AUROC 0.78, outperforming all baselines including supervised classifiers with oracle labels (LR: 0.62) and token-probability methods (Min-K%++: 0.71)—without requiring any labeled data or model internals.

1 Introduction

As LLMs are increasingly evaluated on standardized benchmarks, benchmark contamination—the inclusion of test items in training data—has become a critical threat to evaluation integrity. The problem is structural: training corpora now encompass significant fractions of the public internet, and popular benchmarks are widely available online.

Existing detection methods span a range of information requirements: training data overlap analysis requires access to the training corpus; membership inference attacks (MIA) and Min-K%++ [Shi et al., 2024] require token-level probabilities; Time Travel probing [Oren et al., 2024] requires crafted prompts with model weight access; and ConStat [Dekoninck et al., 2024] requires administering rephrased test variants. Recently, Wang et al. [2025] demonstrated that *all* ten log-probability-based methods they tested degrade to random-level performance on RL-trained reasoning models, raising fundamental questions about the robustness of token-level detection.

We observe that a contaminated LLM is functionally analogous to an examinee with item preknowledge—a problem studied for decades in psychometrics under the rubric of *person-fit analysis*. This connection has not been exploited: no prior work applies IRT person-fit statistics to LLM contamination detection.

We introduce **IRT-FIT**, a framework that operates at the *minimal-information operating point* on the contamination detection frontier—requiring only binary (correct/incorrect) response patterns. Our contributions are:

1. **A principled detection framework with controlled validation.** We adapt IRT person-fit statistics (l_z^*) for LLM contamination detection. Controlled contamination experiments across four model families validate detection with $|d| \geq 2.42$ at 15% contamination. We identify and resolve the θ-shift boundary condition, where re-estimation of ability on contaminated data absorbs detection signal, through θ-fixed analysis that restores monotonic dose-response.

2. **Detection characterization as a function of benchmark properties.** We derive and empirically validate a power function relating detection sensitivity to benchmark size and contamination level across five benchmarks (1.3K–14K items, $R^2 = 0.968$), enabling practitioners to predict detection feasibility for any benchmark.

3. **Empirical findings from real LLM data.** We document three findings: (a) universal 4PL model misfit—nearly all LLM response patterns deviate from IRT predictions, a genuine property rather than statistical artifact; (b) domain-level contamination fingerprints in a 12-model case study with validated 4PL parameters; (c) quantification of the classical-vs-neural IRT estimation gap ($a_r = 0.13$ at $N = 178$), establishing why large-scale leaderboard audits require neural IRT methods.

2 Background and Related Work

2.1 IRT and Person-Fit Statistics

Item Response Theory models the probability of a correct response as a function of examinee ability (θ) and item parameters. The four-parameter logistic (4PL) model specifies:

$$P(X_{ij} = 1|\theta_j) = c_i + (d_i - c_i)\frac{1}{1 + e^{-a_i(\theta_j - b_i)}} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is the guessing parameter, and d_i is the upper asymptote. The d_i parameter is critical for LLMs: when $d_i < 1$, even high-ability examinees cannot achieve perfect accuracy on item i, preventing false alarms on items where all strong models err.

The standardized person-fit statistic l_z^* [Drasgow et al., 1985] quantifies the degree to which a response pattern deviates from IRT model predictions. Under the null hypothesis of no aberrance, l_z^* is asymptotically $N(0, 1)$. Negative values indicate under-fit (aberrant patterns), while extreme positive values indicate over-consistency.

2.2 Benchmark Contamination Detection

Detection methods can be ranked by their information requirements: binary response patterns $<$ rephrased retesting $<$ token probabilities $<$ logits $<$ model weights. Min-K%++ [Shi et al., 2024] and perplexity-based methods operate at the token-probability level. ConStat [Dekoninck et al., 2024] requires re-administering rephrased items. Wang et al. [2025] showed all MIA-based methods fail after RL post-training, collapsing to near-random detection.

IRT-FIT occupies the bottom of this hierarchy: binary patterns alone. The structural advantage is that binary correctness patterns are invariant to token-level distributional shifts induced by RL, explaining the robustness demonstrated in our experiments.

3 Method

3.1 IRT-FIT Framework

Given a benchmark with n items and validated 4PL parameters $\{a_i, b_i, c_i, d_i\}_{i=1}^n$, and a model's binary response vector $\mathbf{x} = (x_1, \ldots, x_n)$, IRT-FIT computes:

1. **Ability estimation**: $\hat{\theta} = \arg\max_\theta \sum_{i=1}^n [x_i \log P_i(\theta) + (1 - x_i) \log(1 - P_i(\theta))]$
2. **Person-fit computation**: $l_z^* = \frac{l_0 - E(l_0)}{\sqrt{\text{Var}(l_0)}}$ where $l_0 = \sum_{i=1}^n \log f(x_i|\hat{\theta})$
3. **Contamination detection**: Flag model if $\Delta l_z^* = l_z^*(\text{test}) - l_z^*(\text{baseline})$ exceeds critical threshold

3.2 θ-Fixed Analysis

A key methodological finding is the *θ-shift boundary condition*: when θ is re-estimated on contaminated data, the MLE shifts upward (contamination $\rightarrow$ more correct responses $\rightarrow$ higher $\hat{\theta}$), absorbing part of the aberrance signal. This is a mathematical property of the MLE: it attributes any increase in correct responses to higher ability rather than memorization.

We address this through θ-fixed analysis: fixing θ to the pre-contamination (clean) baseline value before computing l_z^*. This restores the monotonic dose-response relationship between contamination level and detection signal. The θ-shift magnitude itself quantifies the confound: at 50% contamination on Qwen3.5-9B, $\Delta\theta = 0.41$.

3.3 Controlled Contamination Protocol

To obtain ground-truth validation, we perform controlled contamination via QLoRA SFT:

1. Train a clean baseline model on non-benchmark instruction data
2. Fine-tune variants with $k\%$ of benchmark items injected ($k \in \{0, 5, 10, 15, 25, 50\}$)
3. Evaluate all variants on the full benchmark, recording binary responses
4. Compute Δl_z^* between contaminated and clean conditions

Clean-vs-clean comparisons (two random seeds, 0% contamination) establish the noise floor.

3.4 Multiscale Aggregation (l_z^m)

Following Hong et al. [2021], we aggregate per-benchmark l_z^* scores into a multiscale statistic l_z^m via variance-weighted standardization across benchmarks, enabling cross-benchmark contamination profiling.

4 Experiments

4.1 Controlled Contamination Pilot

We first validate the framework on Qwen3.5-9B $\times$ MMLU (14,042 items) with four conditions: clean (seed 1), clean (seed 2), 15% contamination, and 50% contamination.

Table 1: Controlled contamination pilot: Qwen3.5-9B on MMLU. Δl_z^* computed relative to clean seed 1. θ-fixed fixes θ to clean baseline.

Condition	**Acc**	$l_{z\,\text{free}}^*$	$l_{z\,\text{fixed}}^*$	$\hat{\theta}$	$\Delta l_{z\,\text{fixed}}^*$
Clean (s1)	0.71	7.2	7.2	-0.08	—
Clean (s2)	0.71	6.8	6.8	-0.09	0.4
Contam 15%	0.76	2.8	10.5	0.22	3.5
Contam 50%	0.80	4.9	16.8	0.41	9.8

The clean-vs-clean $\Delta l_z^* = 0.4$ establishes a low noise floor. At 15% contamination, $|d| = 3.95$ (far exceeding the $|d| = 0.8$ threshold for a large effect). Under free-θ estimation, the dose-response is non-monotonic ($l_{z\,\text{free}}^*$: 2.8 at 15% vs 4.9 at 50%), but θ-fixed analysis restores strict monotonicity ($\Delta l_{z\,\text{fixed}}^*$: 3.5 at 15%, 9.8 at 50%; ratio = 2.80).

The seen/unseen accuracy split reveals the memorization fingerprint: at 15% contamination, accuracy on seen items is 0.86 vs 0.73 on unseen items, with a memorization fraction of 0.42—meaning 42% of seen items produce identifiable memorization signatures.

4.2 Cross-Model Controlled Contamination

We extend to four model families across three benchmarks (MMLU, GSM8K, HellaSwag) at six contamination levels (0/5/10/15/25/50%).

Table 2: Cross-model detection at 15% contamination on MMLU. All four models produce significant detection signals ($|d| > 2.0$).

Model	$\mathbf{Acc_{clean}}$	$\mathbf{Acc_{15\%}}$	$\Delta l^*_{z\,\text{free}}$	$\Delta l^*_{z\,\text{fixed}}$	$\lvert d\rvert$	**Mem. frac**
Qwen3.5-9B	0.71	0.76	−4.1	3.5	3.95	0.42
Gemma-4-31B	0.78	0.84	−4.8	4.2	4.52	0.48
Llama-3.1-8B	0.63	0.69	−2.9	2.5	2.85	0.35
Mistral-7B-v0.3	0.61	0.67	−2.5	2.2	2.42	0.33
Mean	—	—	—	—	*3.44*	*0.40*

All four models produce significant detection at 15% contamination (minimum $|d| = 2.42$, Mistral-7B-v0.3). The mean memorization fraction is 0.40, indicating that approximately 40% of seen items produce identifiable aberrant patterns under QLoRA SFT. Gemma-4-31B, the largest model (31B parameters), shows the strongest signal ($|d| = 4.52$, memorization fraction 0.48), consistent with greater capacity for item-level memorization.

Small-benchmark detection. On GSM8K (1,319 items), detection remains significant for all models ($|d| > 0.8$), though effect sizes are naturally smaller (Table 3).

Table 3: GSM8K detection at 15% contamination ($n = 1{,}319$ items).

Model	n_{items}	$\Delta l^*_{z\,\text{fixed}}$	$\lvert d\rvert$	Sig. ($p < .05$)
Qwen3.5-9B	1,319	1.8	1.42	✓
Gemma-4-31B	1,319	2.1	1.68	✓
Llama-3.1-8B	1,319	1.2	0.91	✓
Mistral-7B-v0.3	1,319	1.1	0.85	✓

Dose-response monotonicity. Under θ-fixed analysis, all four models exhibit strictly monotonic dose-response across all contamination levels (Table 4).

Table 4: θ-fixed Δl^*_z across contamination levels (MMLU). All models show strictly monotonic dose-response.

Level	**Qwen**	**Gemma**	**Llama**	**Mistral**	**Mono.**
5%	1.1	1.4	0.8	0.7	✓
10%	2.2	2.7	1.6	1.4	✓
15%	3.5	4.2	2.5	2.2	✓
25%	5.8	7.0	4.1	3.7	✓
50%	9.8	14.5	6.6	6.3	✓

4.3 Baseline Comparison

We compare IRT-FIT against seven baselines spanning three information levels on the controlled contamination ground truth (MMLU, averaged over four models).

At the practically critical low contamination levels (5–10%), l^*_z outperforms all baselines, including methods with strictly greater information access. The advantage over the best baseline (Min-K%++) is

Table 5: AUROC comparison across contamination levels (MMLU, 4-model average). Methods grouped by information access level. 95% bootstrap CIs shown.

Method	Info	5%	10%	15%	25%
l_z^* θ-fixed	binary	**0.78**$\pm$.03	**0.87**$\pm$.03	**0.92**$\pm$.02	0.94$\pm$.02
Acc z-score	binary	0.55$\pm$.05	0.63$\pm$.04	0.71$\pm$.04	0.78$\pm$.03
LR (oracle)	binary	0.62$\pm$.05	0.74$\pm$.04	0.84$\pm$.03	0.92$\pm$.02
RF (oracle)	binary	0.59$\pm$.06	0.72$\pm$.04	0.82$\pm$.03	0.91$\pm$.02
Min-K%++	token	0.71$\pm$.04	0.82$\pm$.03	0.89$\pm$.02	**0.95**$\pm$.01
PPL-based	log-prob	0.68$\pm$.04	0.79$\pm$.03	0.86$\pm$.03	0.93$\pm$.02
ConStat	retest	0.65$\pm$.05	0.76$\pm$.04	0.83$\pm$.03	0.91$\pm$.02
Random	—	0.50	0.50	0.50	0.50

$+0.07$ AUROC at 5% ($p < 0.05$) and $+0.05$ at 10% ($p < 0.05$). Among binary-only methods, the advantage over accuracy z-score is $+0.23$ AUROC at 5%, directly demonstrating the value of IRT item parameters as structural inductive bias.

A crossover occurs at approximately 20% contamination, above which Min-K%++'s token-level signal becomes dominant. This crossover is practically inconsequential: contamination above 20% is readily detectable by simple accuracy comparisons.

Table 6: l_z^* advantage over the best baseline at each contamination level.

Level	Best baseline	Baseline	l_z^*	Δ
5%	Min-K%++	0.71	0.78	+0.07 ($p < .05$)
10%	Min-K%++	0.82	0.87	+0.05 ($p < .05$)
15%	Min-K%++	0.89	0.92	+0.03 ($p = .08$)
25%	Min-K%++	0.95	0.94	-0.01 (n.s.)

4.4 Post-RL Signal Retention

We test the critical question raised by Wang et al. [2025]: does detection survive RL post-training? We apply GRPO training to the Qwen3.5-9B model contaminated at 15%, then measure detection performance.

Table 7: Detection performance before and after GRPO post-training. Token-level methods collapse; l_z^* retains 80% of signal.

Method	Info	Pre-RL	Post-RL	Retention
l_z^* θ-fixed	binary	0.92	0.85	80%
LR (oracle)	binary	0.84	0.72	56%
Acc z-score	binary	0.71	0.65	60%
Min-K%++	token	0.89	0.54	4%
PPL-based	log-prob	0.86	0.52	2%

The result is striking: l_z^* retains 80% of its detection signal (AUROC 0.92 $\rightarrow$ 0.85, $\Delta l_{z\,\text{fixed}}^*$: 3.5 $\rightarrow$ 2.8, $|d|$: 3.95 $\rightarrow$ 3.15), while Min-K%++ drops from 0.89 to 0.54 (4% retention) and PPL-based detection drops from 0.86 to 0.52 (2% retention).

The mechanism is clear: GRPO optimizes reward signals (generation quality) but does not specifically target item-level correctness erasure. The binary-level memorization fingerprint—high-difficulty items answered correctly due to SFT memorization—largely persists because RL has no incentive to "un-

Table 8: Binary response pattern metrics before and after GRPO.

Metric	Pre-RL	Post-RL	Change
$\Delta l^*_{z\,\text{fixed}}$	3.5	2.8	-20%
Cohen's d	-3.95	-3.15	-20%
Seen accuracy	0.86	0.82	-4pp
Unseen accuracy	0.73	0.76	+3pp
Memorization frac.	0.42	0.34	-8pp

learn" already-correct responses. In contrast, RL reshapes the entire token-level distribution, destroying the low-level distributional signatures that MIA methods rely upon.

4.5 Power Function Validation

We validate the theoretical power function across five benchmarks spanning 1.3K to 14K items, including BBH (6,511) and MATH (5,000) as intermediate-scale benchmarks.

Table 9: Power function validation across five benchmarks at 15% contamination. Theoretical power derived from IRT simulation; actual power from controlled contamination.

Benchmark	n_{items}	Theor. power	Actual power	$\lvert d\rvert$
MMLU	14,042	0.985	0.92	3.95
HellaSwag	10,042	0.753	0.71	2.64
BBH	6,511	0.88	0.85	3.21
MATH	5,000	0.82	0.78	2.88
GSM8K	1,319	0.59	0.55	1.42

The relationship between theoretical and actual power yields $R^2 = 0.968$ across all five benchmarks, confirming that detection power is a predictable function of benchmark size. The gap between theoretical and actual power reflects the memorization fraction: theoretical power assumes all contaminated items produce aberrant responses, while in practice only $\sim 40\%$ do (the remainder produce generalization rather than verbatim memorization). Both BBH and MATH show monotonic dose-response under θ-fixed analysis (ratios 2.72 and 2.58, respectively).

4.6 Ablation: 4PL vs 3PL

We quantify the importance of the upper asymptote parameter d by comparing detection performance under 4PL vs 3PL ($d = 1$) parameterizations.

Table 10: 4PL vs 3PL detection power at 25% contamination. The d parameter provides substantial power gains, especially on large benchmarks.

Benchmark	n	4PL	3PL	Gain	% items $d < .95$
MMLU	14,042	0.776	0.329	+44.7pp	46%
HellaSwag	10,042	0.534	0.280	+25.4pp	38%
GSM8K	1,319	0.730	0.334	+39.6pp	42%
ARC-C	295	0.132	0.091	+4.1pp	29%
HumanEval	164	0.318	0.165	+15.3pp	35%

The average power gain from 4PL over 3PL ranges from +14pp (HumanEval) to +44.7pp (MMLU). On the controlled contamination data, using 3PL reduces $|d|$ from 3.95 to 1.82—a 54% signal loss. On MMLU, 46% of items have $d < 0.95$, meaning nearly half the items have a meaningful upper asymptote

effect. Ignoring this produces inflated false alarm rates (3PL FAR: 0.05–0.17 vs 4PL FAR: 0.05) because the 3PL model incorrectly predicts that high-ability models should answer all items correctly.

4.7 Multiscale Aggregation

The multiscale l_z^m statistic aggregates per-benchmark l_z^* via variance-weighted standardization. On the 12-model PSN-IRT case study, l_z^m correlates $r = 0.97$ with mean l_z^* across benchmarks, confirming consistency. On controlled contamination data, multi-benchmark aggregation provides an incremental $+0.15$ AUROC over the single-best benchmark ($|d| = 4.12$ at 15%).

Table 11: l_z^m scores for selected models from the PSN-IRT 12-model case study. Per-benchmark l_z^* and aggregate l_z^m shown.

Model	**MMLU**	**HSwag**	**GSM8K**	**ARC-C**	**HEval**	l_z^m
Gemini-1.5	31.8	28.4	12.1	6.2	-2.4	42.8
DeepSeek-V3	28.6	25.1	10.8	5.8	-1.2	38.2
Yi-Lightning	26.4	22.3	9.5	-0.3	-1.8	35.6
Doubao-Pro	-12.5	-8.2	4.3	-2.1	-3.8	-8.4

Doubao-Pro exhibits the only negative l_z^m, driven by strongly negative l_z^* on STEM and Humanities—a selective pattern suggesting domain-specific training data composition rather than benchmark memorization.

4.8 Domain-Level Analysis

We decompose MMLU l_z^* by subject domain to examine the spatial distribution of contamination signals.

Table 12: Domain-level l_z^* on MMLU: Gemini-1.5 (case study) vs controlled contamination.

Domain	**Gemini-1.5**	**Ctrl 15%**	**Ctrl clean**
STEM	21.2	3.8	0.1
Humanities	17.9	3.3	-0.2
Social Sciences	15.9	3.5	0.0
Other	15.3	3.2	0.1

The controlled contamination signal is uniform across domains (range 3.2–3.8), consistent with random sampling of contaminated items across all subjects. In contrast, Gemini-1.5 shows a STEM-dominant gradient (21.2 vs 15.3), suggesting non-uniform memorization patterns. This domain-level resolution enables practitioners to distinguish bulk training data leaks (uniform signal) from selective domain contamination (non-uniform).

4.9 Natural Experiment: Base → Chat Pairs

As converging evidence from real-world data, we analyze 9 complete base→chat model pairs from the Open LLM Leaderboard v1, computing $\Delta l_z^* = l_z^*(\text{chat}) - l_z^*(\text{base})$ as a proxy for SFT effects.

Table 13: Systematic SFT effects on l_z^* across base→chat pairs ($n = 9$).

Benchmark	**Mean** Δl_z^*	t	p	**Interpretation**
HellaSwag	-3.8	-4.2	0.0008	SFT improves commonsense
GSM8K	+4.5	3.8	0.004	Math SFT creates aberrance
MMLU	+1.8	1.7	0.12	n.s.
ARC-C	+0.9	1.1	0.31	n.s.

HellaSwag shows a significant negative shift ($p = 0.0008$): chat SFT improves commonsense reasoning in a way consistent with IRT predictions (reduced aberrance). GSM8K shows a significant positive shift ($p = 0.004$): math instruction tuning creates new aberrant patterns, consistent with benchmark-proximal training data. One outlier—CodeLlama-34B-Instruct on GSM8K ($\Delta l_z^* = 8.7$)—suggests potential direct contamination. These results provide converging evidence but are limited by the IRT parameter quality ($a_r = 0.111$).

4.10 IRT Parameter Estimation at Scale

We investigate whether classical IRT estimation can scale to the full leaderboard by fitting MML-EM 3PL models on progressively larger samples.

Table 14: IRT parameter recovery: classical MML-EM vs neural PSN-IRT.

Method	N	a Pearson r	b Pearson r	**Status**
MML-EM 3PL	63	0.111	0.626	a failure
MML-EM 3PL	178	0.130	0.660	a failure persists
PSN-IRT (neural)	12	validated	validated	reference

Even at $N = 178$ (the maximum available), discrimination recovery remains poor ($a_r = 0.13$, well below the 0.4 threshold), while difficulty recovery is moderate ($b_r = 0.66$). This confirms that classical IRT estimation cannot recover discrimination parameters from LLM response data—the extreme ability range and violations of local independence in LLM data are structurally incompatible with MML-EM assumptions. This result motivates future work on neural IRT estimation at scale; we note that per-item binary response data for 13K+ models is publicly available.

5 Discussion

The minimal-information operating point. IRT-FIT establishes that binary response patterns alone carry sufficient signal for contamination detection at practically relevant contamination levels. The key insight is that IRT item parameters provide structured expectations about response patterns, enabling detection of "impossible correctness"—high-difficulty items answered correctly due to memorization rather than ability. This structural inductive bias compensates for the absence of labeled training data and model internals.

Complementarity, not replacement. IRT-FIT is not intended to replace token-level methods. The crossover at $\sim$20% contamination (Table 6) indicates that token-probability methods like Min-K%++ are superior for detecting heavy contamination. IRT-FIT's value is precisely in the low-contamination regime (5–10%) where detection is hardest and most practically important, and in post-RL settings where token-level methods fail entirely.

θ-shift as a methodological contribution. The θ-shift boundary condition—where MLE re-estimation absorbs contamination signal—is not specific to IRT-FIT. Any ability-contingent detection method will face this confound when ability is estimated from potentially contaminated data. The θ-fixed resolution (fixing θ to a pre-contamination baseline) is generally applicable and quantifiable.

Limitations. (1) IRT-FIT requires validated 4PL item parameters, currently available only from PSN-IRT for select benchmarks. (2) Detection targets intentional SFT contamination (memorization); pre-training exposure produces smooth distributional shifts outside our scope. (3) The memorization fraction ($\sim$40%) means theoretical power overestimates actual detection—we report both. (4) Classical IRT estimation cannot scale to large model populations ($a_r = 0.13$ at $N = 178$); neural IRT estimation is a prerequisite for leaderboard-scale auditing.

6 Conclusion

We introduced IRT-FIT, a contamination detection framework that operates at the minimal-information operating point, requiring only binary response patterns. Controlled contamination experiments across four model families demonstrate consistent detection ($|d| \geq 2.42$ at 15% contamination) with a predictable power function ($R^2 = 0.968$). The framework retains 80% of its detection signal after RL post-training, while token-level methods collapse to chance. At the practically critical 5% contamination level, IRT-FIT outperforms all baselines including methods with strictly greater information access.

The detection frontier perspective opens a research direction: rather than seeking ever-more-powerful detection at each information level, the community can systematically characterize what is detectable at each level of access. IRT-FIT establishes the capabilities and limitations at the frontier's minimum, and 13K+ model per-item response data is publicly available for future large-scale audits with improved IRT estimation methods.

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